

Shared evidence | Plant transport and air movement

Plant transport: follow the water and sugar

Name _____ Date _____

Constructed potometer data: same shoot, capillary, light and 20°C. Each value is bubble movement over 5 minutes after adjustment.

Air condition	Three movements(mm)	Time for each(min)
Still air	12,11,13	5
Moving air	30,28,32	5

Roots and xylem

Root hairs provide a large surface. Water can enter by osmosis; mineral ions may require active transport. Hollow, lignified xylem conducting cells carry water and mineral ions.

Leaf and air

Water evaporates from moist internal surfaces. Vapour diffuses through stomata. Moving air may remove humid air around the leaf.

Phloem

Living tissue moves dissolved sucrose from source to sink. A source leaf may supply roots below and a growing bud above using different tubes.

Potometer limit

Bubble movement indicates water uptake. Some water is used or retained; leaks can affect readings. Uptake does not exactly equal transpiration.

W1. Calculate rates. W2. Explain the air effect. W3. Correct the exact-evaporation claim.

Supported route | Trace and calculate

Word bank: water/mineral ions; sucrose; xylem; phloem; evaporation; diffusion; uptake.

S1. Draw two labelled arrows: roots → leaf for water/mineral ions, and leaf → a sink for sucrose. Name each tissue.

S2. Describe evaporation inside the leaf and diffusion out of the stoma.

S3. Calculate the mean movement and rate for still and moving air.

S4. Explain why the faster bubble is evidence about uptake, not an exact transpiration measurement.

Standard route | Write the transport report

Use the source cards and repeat table.

M1. Compare xylem and phloem by transported material, cell structure and direction/source-to-sink role.

M2. Explain how water moves from root to surrounding air, including osmosis, xylem, evaporation and diffusion.

M3. Calculate both mean movements and rates. Describe the difference using units.

M4. Explain the air-movement result and give two requirements for a fair, valid comparison.

Stretch route | From distance to volume

Optional calculation. For this constructed capillary, cross-sectional area=0.5 mm².
Volume=area×distance. Use the shared mean movements.

T1. Calculate mean water-uptake volume and volume per minute in both conditions. Give mm³ and mm³/min.

T2. Why would comparing distance rates in tubes with different diameters be misleading?

T3. A shoot wilts and its stomata close during a stronger fan trial. Must uptake keep rising? Explain a competing effect.

T4. Propose a check for an apparatus error and a separate improvement to the biological comparison.

Exit and reflection | Follow the material

Use labels and short reasons.

E1. What tissue carries sucrose from source to sink?

E2. Calculate 20 mm in 5 minutes, with a unit.

E3. Give one reason uptake need not equal exact transpiration.

L1. What did your audience question? Improve one arrow, label or claim.

Next useful step

- A. Distinguish the tissues.
- B. Explain evaporation then diffusion.
- C. Calculate a rate.
- D. State the proxy limit.